

$$\begin{aligned}
 &= (A \cup B) - ((A \cup B) \cap ((B \cap \bar{A}) \cup (B \cap \bar{B}))) \quad (2nd \text{ Distributive Law}) \\
 &= (A \cup B) - ((A \cup B) \cap ((B \cap \bar{A}) \cup \emptyset)) \quad (2nd \text{ Complement Law}) \\
 &= (A \cup B) - ((A \cup B) \cap (B \cap \bar{A})) \quad (1st \text{ Identity Law}) \\
 &= (A \cup B) \cap \overline{(A \cup B) \cap (B \cap \bar{A})} \quad (\text{Prob 21(a)}) \\
 &= (A \cup B) \cap (\overline{A \cup B} \cup \overline{B \cap \bar{A}}) \quad (1st \text{ De Morgan's Law}) \\
 &= (A \cup B) \cap (\overline{A \cup B} \cup (B \cap \bar{A})) \quad (2nd \text{ De Morgan's Law}) \\
 &= (A \cup B) \cap ((\bar{A} \cap \bar{B}) \cup (B \cap \bar{A})) \quad (1st \text{ De Morgan's Law}) \\
 &= (A \cup B) \cap ((\bar{A} \cap \bar{B}) \cup (\bar{B} \cap \bar{A})) \quad (1st \text{ De Morgan's Law}) \\
 &= (A \cup B) \cap ((\bar{A} \cap \bar{B}) \cup (\bar{B} \cap \bar{A})) \quad (\text{Complementation Law}) \\
 &= (A \cup B) \cap ((\bar{A} \cap \bar{B}) \cup (\bar{B} \cap \bar{A})) \quad (2nd \text{ Distributive Law}) \\
 &= ((A \cup B) \cap (\bar{A} \cap \bar{B})) \cup ((A \cup B) \cap (\bar{B} \cap \bar{A})) \quad (2nd \text{ De Morgan's Law}) \\
 &= ((A \cup B) \cap (\bar{A} \cap \bar{B})) \cup ((A \cup B) \cap (\bar{B} \cap \bar{A})) \quad (2nd \text{ Complement Law}) \\
 &= \emptyset \cup ((A \cup B) \cap (\bar{B} \cap \bar{A})) \quad (1st \text{ Commutative Law}) \\
 &= ((A \cup B) \cap (\bar{B} \cap \bar{A})) \cup \emptyset \quad (1st \text{ Commutative Law}) \\
 &= ((A \cup B) \cap (\bar{B} \cap \bar{A})) \quad (2nd \text{ Identity Law}) \\
 &= ((A \cup B) \cap (\bar{B} \cap \bar{A})) \quad (1st \text{ Commutative Law}) \\
 &= (A \cup B) \cap (\bar{A} \cap \bar{B}) \quad (1st \text{ Commutative Law}) \\
 &= A \cup (B \cap \bar{B}) \quad (1st \text{ Distributive Law}) \\
 &= A \cup \emptyset \quad (2nd \text{ Complement Law}) \\
 &= A \quad (2nd \text{ Identity Law}) \quad \square
 \end{aligned}$$

45) What can you say about the sets A and B if $A \oplus B = A$?

Solution: $A \oplus B = (A \cup B) - (A \cap B) \quad (\text{Prob 41})$
 $= (A \cup B) \cap \overline{A \cap B} \quad (\text{Prob 21(a)})$
 $= (A \cup B) \cap (\bar{A} \cup \bar{B}) \quad (1st \text{ De Morgan's Law})$

Given, $(A \cup B) \cap (\bar{A} \cup \bar{B}) = A$
 $\therefore (A \cup B) \cap (\bar{A} \cup \bar{B}) \subseteq A$ and $A \subseteq (A \cup B) \cap (\bar{A} \cup \bar{B})$.

Let, x be an arbitrary element of A, i.e. $x \in A$. By Defn. 3 from Sec-2.1, $x \in (A \cup B) \cap (\bar{A} \cup \bar{B})$. By Defn. 2, $x \in (A \cup B) \wedge x \in (\bar{A} \cup \bar{B})$. By Defn. of AND, $x \in (A \cup B)$ and $x \in (\bar{A} \cup \bar{B})$. By Defn. 1, $x \in A \vee x \in B$. By Defn. 5, $x \notin A \vee x \in \bar{B}$. By Defn. of not belongs to, $\neg(x \in A) \vee x \in \bar{B}$. Since, $x \in A$ by disjunctive Syllogism, $x \in \bar{B}$. By Defn. 5, $x \notin B$. By Defn. of not belongs to, $\neg(x \in B)$. By Defn. of negation,